

Short Communication

Emergence of a highly colistin-resistant *Aeromonas jandaei* clinical isolate harbouring four genes encoding phosphoethanolamine transferases in NepalTomoki Komeda^a, Shovita Shrestha^b, Jatan B. Sherchan^c, Mari Tohya^a, Tomomi Hishinuma^a, Jeevan B. Sherchand^b, Tatsuya Tada^{a,*}, Teruo Kirikae^a^a Department of Microbiology, Juntendo University School of Medicine, Tokyo, Japan^b Department of Microbiology, Institute of Medicine, Tribhuvan University, Kathmandu, Nepal^c Department of Clinical Microbiology, Kathmandu University School of Medical Sciences, Dhulikhel, Nepal

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ABSTRACT

Objectives: This study aimed to describe a clinical isolate of *Aeromonas jandaei* (*A. jandaei*) in Nepal that harboured four types of genes encoding phosphoethanolamine transferases.**Methods:** An isolate of colistin-resistant *A. jandaei* was obtained from a blood sample of an inpatient in a hospital in Nepal, and its complete genome sequence was determined. *Escherichia coli* (*E. coli*) and *Aeromonas hydrophila* (*A. hydrophila*) transformants expressing genes encoding novel phosphoethanolamine transferase variants were constructed and colistin-susceptibility profiles were determined.**Results:** The isolate harboured four genes encoding phosphoethanolamine transferases on the chromosome, which were designated *eptAv3.2*, *eptAv3.3*, *eptAv3.4* and *eptAv7.2*. The amino acid sequences of *EptAv3.2*, *3.3* and *3.4* were > 80% identical to MCR-3.1, and that of *EptAv7.2* was > 79% identical to MCR-7.1. *E. coli* expressing *eptAv3.2*, *3.3* and *3.4* showed reduced susceptibility to colistin, whereas *E. coli* expressing *eptAv7.2* did not. In contrast, *A. hydrophila* expressing *eptAv7.2* showed reduced susceptibility to colistin, whereas *A. hydrophila* expressing *eptAv3.2*, *3.3* and *3.4* did not; *eptAv3.3* and *3.4* formed a tandem structure. The genomic environments surrounding *eptAv3.2*, *3.3* and *3.4* were similar to *Aeromonas veronii* obtained from the effluent of a treatment plant in Japan in 2018. The genomic environment surrounding *eptAv7.2* was similar to that of *A. jandaei* obtained from a chicken in the USA in 2019.**Conclusions:** The highly colistin-resistant *A. jandaei* clinical isolate harboured four chromosomal genes encoding phosphoethanolamine transferases, suggesting that *Aeromonas* spp. harbouring *eptAv* genes with strong similarities to *mcr-3* and *mcr-7* are emerging in medical settings as well as environments.

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1. Introduction

The emergence of Gram-negative bacteria that are resistant to colistin, which is a 'last-resort' antibiotic, has become a serious problem worldwide [1]. Colistin is a drug used to treat carbapenem-resistant Enterobacteriaceae infections. Due to its high toxicity, use of colistin was avoided in humans for a long time, but it has recently been used to treat multidrug-resistant infections [2]. In 2015, colistin-resistant *Escherichia coli* and *Klebsiella pneu-*

moniae strains harbouring plasmids encoding *mcr-1* were detected in humans and livestock from China, leading to the discovery of worldwide dissemination of *mcr-1*-positive strains of Enterobacteriaceae [3,4]. To date, various types of plasmid-mediated mobilised colistin-resistance *mcr* genes have been identified, including *mcr-1* to *mcr-10* [5,6].

Aeromonas jandaei (*A. jandaei*) was originally isolated from clinical samples – including blood, wounds and diarrheal stools – in several hospitals in the USA in 1991 [7]. To date, most strains of *A. jandaei* have been isolated from environmental samples and animal-origin samples [8]. A strain of this bacterium isolated from retail fish samples in Guangdong Province, China, was found to harbour two genes – *eptAv-3* and *eptAv-7* – that were similar to *mcr-3* and *mcr-7*, respectively. These findings suggested that

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Aeromonas spp. were reservoirs for the spread of colistin-resistant bacteria [8].

This study describes a patient in Nepal with isolation of a highly colistin-resistant strain of *A. jandaei* harbouring four chromosomal genes encoding phosphoethanolamine transferases.

2. Materials and methods

2.1. Bacterial strains and drug susceptibility

The *A. jandaei* strain JUNP479 was isolated from a blood sample obtained from an inpatient at Tribhuvan University Teaching Hospital, Nepal, in October 2019. *E. coli* DH5 α (Takara Bio, Shiga, Japan) and *Aeromonas hydrophila* (*A. hydrophila*) IOMTU903 were used as hosts for recombinant plasmids harbouring *eptAv3.2*, *eptAv3.3*, *eptAv3.4* and *eptAv7.2*. Bacteria were identified by mass spectrometry using Bruker MALDI Biotyper System (Bruker Daltonics, Germany) and 16S rRNA sequencing using ABI 3500xL Genetic Analyzer (Applied Biosystems, Foster City, CA). The MICs of the antimicrobial agents were determined using the microdilution method, as described in CLSI 2020 guidelines [9].

2.2. Complete genome sequencing

The whole genome of the isolate was extracted using DNeasy Blood and Tissue kits (Qiagen, Tokyo, Japan) for short-read sequencing using MiSeq, and QIAGEN Genomic-tip 100/G and Genomic DNA Buffer Set (Qiagen, Tokyo, Japan) for long-read sequencing using MinION (Oxford Nanopore Technologies, Oxford, UK). Raw reads from MiSeq and MinION were assembled using Unicycler v0.4.7 [10]. Genome homology was searched using Basic Local Alignment Search Tool (BLAST) (https://blast.ncbi.nlm.nih.gov/Blast.cgi?PROGRAM=blastn&PAGE_TYPE=BlastSearch&LINK_LOC=blasthome). Genomic environments surrounding genes encoding phosphoethanolamine transferases were compared using in silico Molecular Cloning (In Silico Biology, Inc, Yokohama, Japan).

Phylogenetic dendrograms of MCR and EptAv variants were created using the CLUSTAL OMEGA program (<https://www.ebi.ac.uk/Tools/msa/clustalo/>).

2.3. *E. coli* and *A. hydrophila* transformants expressing phosphoethanolamine transferases

The open reading frames (ORFs) and promoter regions of *eptAv3.2*, *eptAv3.3*, *eptAv3.4* and *eptAv7.2* in *A. jandaei* JUNP479 were amplified using primer sets (Supplementary Table). The PCR products were digested with EcoI and PstI and ligated into the plasmid pHSG398 (Takara Bio, Shiga, Japan) [11]. The plasmids were used to transform *E. coli* DH5 α and *A. hydrophila* IOMTU903, and transformants were selected on LB agar plates containing 30 mg/L chloramphenicol. The susceptibilities of these transformants to colistin were assayed.

2.4. Real-time PCR for genes encoding phosphoethanolamine transferases

Total RNAs from *E. coli* DH5 α and *A. hydrophila* IOMTU903 expressing *eptAv3.2*, *eptAv3.3*, *eptAv3.4* and *eptAv7.2* were extracted using RNeasy mini kit (Qiagen, Tokyo, Japan) and reverse-transcribed to cDNA using the PrimeScriptTM reverse transcriptase (Takara Bio, Shiga, Japan). Quantitative real-time PCR was performed using primer sets (Supplementary Table) and SYBR Premix Ex Taq II kit (Takara, Shiga Japan). The signals were detected by the QuantStudio[®] 3 real-time PCR system (Thermo Fisher Scientific, MA, USA). The expressions of the genes were normalized to 16S rRNA. All samples were tested in triplicate in three independent experiments.

Table 1

Colistin susceptibility profiles of *Aeromonas jandaei* JUNP479 harbouring *eptAv3.2*, *eptAv3.3*, *eptAv3.4* and *eptAv7.2* and their transformants.

Strain	MIC (mg/L) of colistin
<i>A. jandaei</i> JUNP479	128
<i>E. coli</i> DH5 α +pHSG398- <i>eptAv3.2</i>	2
<i>E. coli</i> DH5 α +pHSG398- <i>eptAv3.3</i>	1
<i>E. coli</i> DH5 α +pHSG398- <i>eptAv3.4</i>	0.5
<i>E. coli</i> DH5 α +pHSG398- <i>eptAv7.2</i>	0.125
<i>E. coli</i> DH5 α +pHSG398	0.125
<i>A. hydrophila</i> +pHSG398- <i>eptAv3.2</i>	0.063
<i>A. hydrophila</i> +pHSG398- <i>eptAv3.3</i>	0.063
<i>A. hydrophila</i> +pHSG398- <i>eptAv3.4</i>	0.063
<i>A. hydrophila</i> +pHSG398- <i>eptAv7.2</i>	4
<i>A. hydrophila</i> +pHSG398	0.063

2.5. GenBank accession number

The complete genome sequence of *A. jandaei* JUNP479 determined in this study was deposited in GenBank under accession number AP024466.

3. Results

3.1. Drug susceptibilities and drug-resistant genes

Aeromonas jandaei JUNP479 showed resistance to ceftriaxone (MIC 8 mg/L) and intermediate resistance to imipenem (MIC 2 mg/L), but was susceptible to ceftazidime (MIC 0.5 mg/L), meropenem (MIC $\leq$ 0.25 mg/L), aztreonam (MIC $\leq$ 0.25 mg/L), amikacin (MIC 4 mg/L), gentamicin (MIC $\leq$ 0.25 mg/L), ciprofloxacin (MIC $\leq$ 0.25 mg/L) and tetracycline (MIC 1 mg/L). The MICs of colistin and tigecycline were 128 mg/L and 64 mg/L, respectively.

The isolate was found to harbour *bla*_{FOX}, encoding a class C β -lactamase, and four novel variants of genes encoding phosphoethanolamine transferases, designated *eptAv3.2*, *eptAv3.3*, *eptAv3.4* and *eptAv7.2*. The amino acid sequences of EptAv3.2, EptAv3.3 and EptAv3.4 were 80.7%, 95.7% and 84.7% identical to that of MCR-3.1, respectively, and 98.1%, 80.4% and 84.0% identical to that of EptAv3, respectively. The amino acid sequence of EptAv7.2 was 79.9% and 97.6% identical to the sequences of MCR-7.1 and EptAv7, respectively. The variants *eptAv3.2*, *eptAv3.3*, *eptAv3.4* showed 80.7%, 95.7% and 84.7% amino acid identity to MCR-3.1, respectively, and the amino acid sequence of *eptAv7.2* was 79.9% identical to that of MCR-7.1.

3.2. Transformants expressing novel phosphoethanolamine transferase variants

Escherichia coli DH5 α expressing *eptAv3.2*, *eptAv3.3* and *eptAv3.4* showed reduced susceptibility to colistin, whereas DH5 α expressing *eptAv7.2* did not (Table 1). In contrast, *A. hydrophila* IOMTU903 expressing *eptAv7.2* showed reduced susceptibility to colistin, whereas IOMTU903 expressing *eptAv3.2*, *eptAv3.3* and *eptAv3.4* did not.

3.3. Expressions of *eptAv3* and *eptAv7* variants in *E. coli* and *A. hydrophila* transformants

The expression of plasmid pHSG398-cloned gene in *E. coli* was two-fold higher than that in *A. hydrophila* (Supplementary Figure 1A). The three *eptAv3* variants in *E. coli* DH5 α were expressed two-fold higher than *eptAv7.2*, whereas the three *eptAv3* variants in *A. hydrophila* were expressed 10-fold lower than *eptAv7.2* (Supplementary Figures 1B and 1C).

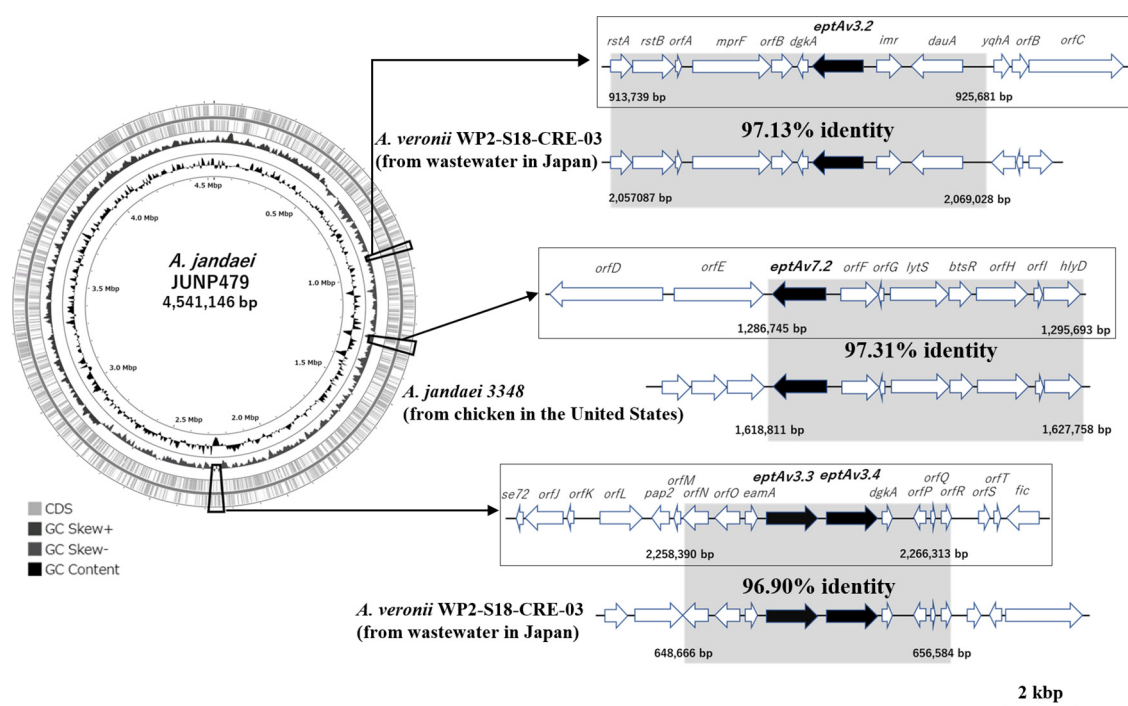


Figure 1. Genomic environments surrounding *eptAv3.2*, *eptAv3.3*, *eptAv3.4* and *eptAv7.2* in *Aeromonas jandaei* JUNP479.

All four genes encoding phosphoethanolamine transferases were contained in the chromosome. The genomic environments surrounding *eptAv3.2*, *eptAv3.3* and *eptAv3.4* were similar to those in *Aeromonas veronii* (A. veronii) WP2-S18-CRE-03 (accession no. AP021940), whereas that surrounding *eptAv7.2* was similar to that in *Aeromonas jandaei* (A. jandaei) 3348 (accession no. CP043321).

3.4. Genomic environments surrounding genes encoding phosphoethanolamine transferases

Complete genome sequencing revealed that *A. jandaei* JUNP479 has a chromosome of size 4 534 922 base-pairs (bp), with 49.93% guanine-cytosine content and a plasmid the size of 6224 bp. The sequencing achieved more than 600-fold coverage of the genome. All four genes encoding phosphoethanolamine transferases were located on the chromosome, with *eptAv3.3* and *eptAv3.4* forming a tandem structure (Figure 1). The genomic environment surrounding *eptAv3.2* was *rstA-rstB-orfA-mprF-vff-dgkA1-eptAv3.2-imr-dauA-yqhA-orfB-orfC*, with the *orfA*, *B* and *C* genes encoding hypothetical proteins (Figure 1). The structure *rstA-rstB-orfA-mprF-vff-dgkA1-eptAv3.2-imr-dauA* (nt 913 739 to nt 925 681) was identical to that in *A. veronii* WP2-S18-CRE-03 (nt 2 069 028 to nt 2 057 087) obtained from the effluent of a treatment plant in Japan in 2018 (GenBank accession no. AP021940). The genomic environment surrounding *eptAv7.2* was *orfD-orfE-eptAv7.2-orfF-orfG-lytS-btsR-orfH-orfI-hlyD*, with the *orfD*, *G* and *H* genes encoding hypothetical proteins; the *orfE* gene encoding beta-N-acetylhexosaminidase, the *orfF* gene encoding a phosphodiesterase, and the *orfI* gene encoding a membrane protein (Figure 1). The structure, *eptAv7.2-orfF-orfG-lytS-btsR-orfH-orfI-hlyD* (nt 1 286 745 to nt 1 295 693), was identical to that in *A. jandaei* 3348 (nt 1 618 811 to nt 1 627 758), which was isolated from a chicken in the USA in 2019 (GenBank accession no. CP043321). The genomic environment surrounding *eptAv3.3* and *eptAv3.4* was *se72-orfJ-orfK-orfL-pap2-orfM-orfN-orfO-eamA-eptAv3.3-eptAv3.4-dgkA-orfP-orfQ-orfR-orfS-orfT-fic*, with the *orfJ* gene encoding a reverse transcriptase; the *orfK*, *L*, *M*, *P*, *Q*, *R* and *T* genes encoding hypothetical proteins; the *orfN* and *O* genes encoding membrane proteins; and the *orfS* gene encoding H-NS histone protein (Figure 1). The structure *eptAv3.3-eptAv3.4-dgkA-orfP-orfQ-orfR* (nt 2 258 390 to nt 2 266 313) was identical to that in *A. veronii* WP2-S18-CRE-03 (nt 648 666 to nt

656 584), which was isolated from the effluent of a treatment plant in Japan in 2018 (GenBank accession no. AP021940).

Mobile genetic elements including transposase genes and insertion sequences were not detected in the 40 000 bp around the *eptA* genes on the chromosome of *A. jandaei* JUNP479.

Phylogenetic dendrograms of MCR and EptAv variants revealed that MCR-3.1 in *E. coli* (NG_055505) showed close phylogenetic distance to MCR-3 family and EptAv3.3 in *Aeromonas* spp., whereas MCR-7.1 in *Klebsiella pneumoniae* (NG_056413) showed considerable phylogenetic distance from MCR-7 family and EptAv7 variants in *Aeromonas* spp. (Supplementary Figure 2).

4. Discussion

It is believed that this is the first report of a highly colistin-resistant *A. jandaei* clinical isolate harbouring four genes encoding phosphoethanolamine transferases in Nepal. Previous studies have reported that genes encoding phosphoethanolamine transferases were detected in environmental, animal-origin and clinical samples of *Aeromonas* species – including *A. allosaccharophila*, *A. bivalvium*, *A. caviae*, *A. hydrophila*, *A. jandaei*, *A. media*, *A. salmonicida* and *A. veronii* – isolated from water, chicken, pork, retail fish and/or clinical samples [8,12–16]. These strains harboured both *mcr-3*-like and/or *mcr-7*-like genes in the chromosomes [8,12–16].

The promoter regions upstream of the *eptA* genes have been associated with phosphoethanolamine transferase expression in *Aeromonas* species or *E. coli*. In the present study of *A. jandaei* JUNP479, the *eptAv3* promoter would control expression in *E. coli*, whereas *eptAv7* would control expression in *A. hydrophila* (Supplementary Figure 1).

Aeromonas species may be reservoirs of specific phosphoethanolamine transferases encoding genes such as *mcr-3* and *mcr-7*. *Aeromonas* spp. harbouring *eptAv* genes with strong similarities to *mcr-3* and *mcr-7* seem to be spreading in medical and

environmental fields worldwide. The amino acid sequences of four *eptA* variant genes, including *eptAv3.2*, *eptAv3.3*, *eptAv3.4* and *eptAv7.2*, showed strong similarities to those in other *Aeromonas* spp. such as *A. allosaccharophila*, *A. finlandensis*, *A. lacusus*, *A. sobria*, *A. veronii* and *A. jandaei* (Supplementary Figure 2). In particular, *mcr-3* has frequently been detected in Enterobacteriaceae as well as *Aeromonas* spp. clinical isolates, compared with *mcr-7* [5]. It is necessary to survey *mcr-3* and *mcr-7* harbouring isolates of *Aeromonas* and Enterobacteriaceae in environmental and medical fields.

5. Conclusions

A highly colistin-resistant *A. jandaei* clinical isolate with four genes encoding phosphoethanolamine transferases was isolated in a medical setting in Nepal. Three of the four genes – *eptAv3.2*, *3.3* and *3.4* – were functional in *E. coli*, whereas one, *eptAv7.2*, was functional in *A. hydrophila*. These results suggest that *Aeromonas* spp. harbouring *eptAv* genes with strong similarities to *mcr-3* and *mcr-7* are emerging in environmental and medical fields.

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Competing Interests

No conflict of interest.

Ethical Approval

This study was approved by the Ethical Committee, Institute of Medicine, Tribhuvan University [219 (6-11-E)2/075/076], and the Biosafety Committee, Juntendo University (BSL2/29-1).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.ijantimicag.2022.106544](https://doi.org/10.1016/j.ijantimicag.2022.106544).

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